

# Spatial Inventory: An AI-Enhanced Pipeline for Authoring and Retrieving AR Documentation

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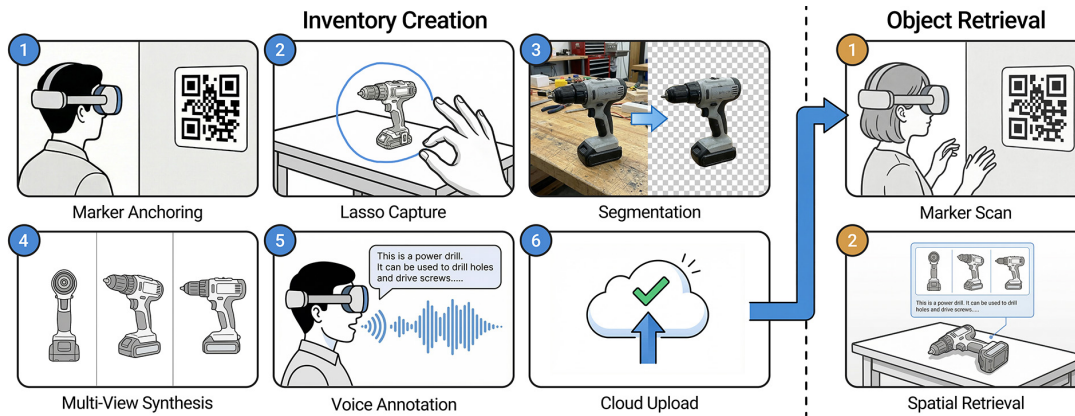


Figure 1: System overview of Spatial Inventory. *Left (Inventory Creation)*: (1) Marker-based spatial anchoring establishes a persistent reference frame; (2) freehand lasso gesture captures target object; (3) SAM2 removes background for clearer documentation; (4) multi-view synthesis generates additional perspectives; (5) voice annotation with transcription and optional AI refinement; (6) cloud upload. *Right (Object Retrieval)*: (1) Marker scan fetches session data; (2) AR info panels appear at stored 3D positions.

## ABSTRACT

Shared workspaces such as makerlabs accumulate diverse tools, yet knowledge of where items reside and how they are used often remains tacit, transferring inefficiently to newcomers. Existing AR systems either operate reactively or require pre-scanned environments and technical expertise for content authoring. We present Spatial Inventory, an AR system for head-mounted displays that enables staff to proactively create spatially-anchored documentation. Users select objects via freehand lasso gestures, optionally apply AI-enhanced background removal and multi-view synthesis, add voice annotations, and upload to cloud storage. Newcomers retrieve these annotations at their original 3D positions using marker-based spatial anchoring.

**Index Terms:** Augmented Reality, Spatial Annotation, Knowledge Transfer, Object Segmentation.

## 1 INTRODUCTION

Newcomers to shared workspaces such as makerlabs, workshops, and warehouses often face a recurring challenge: locating tools and learning how equipment is used. Experienced staff possess valuable implicit knowledge about where items are stored, what each tool does, and which equipment requires special handling. Yet this knowledge often remains undocumented and transfers inefficiently to new members through ad-hoc verbal explanations or shadowing sessions.

Augmented Reality (AR) offers a promising solution: spatially-anchored annotations that new members can retrieve at the exact locations where tools reside. However, existing approaches such

as traditional documentation lack spatial context, making it difficult for newcomers to connect instructions with physical locations. AR systems for knowledge transfer often focus on *reactive* assistance, where experts respond in real-time when novices encounter problems, rather than enabling *proactive* documentation before issues arise. Furthermore, AR authoring tools typically require pre-scanned environments, pre-authored 3D models, or technical expertise [3, 2], creating barriers for non-technical staff who hold the knowledge but lack means to capture it.

We present **Spatial Inventory**, an AR system that enables experienced staff to create spatially-anchored documentation for later retrieval by newcomers. Users select objects of interest via freehand lasso gestures, optionally refine captures through segmentation, and add voice annotations describing the item's purpose and usage. Newcomers entering the same space can retrieve these annotations at their original 3D positions using a Marker-based spatial anchor. The system aims to lower barriers by supporting ad-hoc annotation without pre-scanning or pre-authored 3D content.

We contribute an in-situ pipeline for ad-hoc AR documentation, coupling promptable segmentation and multi-view synthesis, plus a proactive-asynchronous workflow.

## 2 RELATED WORK

**AR for Training and Knowledge Transfer.** MR collaboration research has predominantly focused on synchronous scenarios; Sereno et al. [8] found only 3 papers on asynchronous collaboration compared to 47 on synchronous. ProcessAR [1] enables experts to author procedural AR instructions in-situ, but requires pre-trained object detection models. Existing systems focus on reactive assistance (novice encounters problem → expert responds) rather than proactive documentation.

**AR Authoring Systems.** Nebeling and Speicher [3] identified a persistent gap in AR authoring: most tools require pre-authored 3D content or technical expertise. ScalAR [4] supports semantically

adaptive AR but requires authoring in VR rather than in-situ. Even ImagineAR [2], which combines SAM-based scene understanding with LLM-driven generation, depends on pre-scanned locations processed offline.

### 3 TECHNIQUE

#### 3.1 Inventory Creation Workflow

Spatial Inventory implements a six-stage pipeline for creating spatially-anchored inventory items (Figure 1, left).

**Stage 1: Spatial Anchoring.** Users scan a printed marker to establish a spatial reference frame. All subsequent positions are stored relative to this anchor. This enables cross-session retrieval: when users return and scan the same marker, previously documented items reappear at their original locations.

**Stage 2: Lasso Selection and Capture.** Users pinch to begin drawing a freehand loop around the target object; a second pinch confirms the capture, extracting the enclosed region from the passthrough camera (1280×1280). The system simultaneously casts depth rays toward the lasso region to estimate 3D position.

**Stage 3: Optional SAM2 Background Removal.** Users can tap on the preview panel to indicate points of interest on the target object, then pinch to confirm and send these prompts to SAM2 [7]. The model returns an object mask and background-removed cutout for clearer documentation.

**Stage 4: Optional Multi-View Synthesis.** To supplement limited passthrough camera resolution when single-view captures may not fully convey an object’s appearance, users can generate additional perspectives via Nano Banana Pro [6]. These synthesized views provide approximate visual references for newcomers, though outputs may lack precision or contain hallucinations.

**Stage 5: Voice Annotation.** Users record speech describing the item. The system transcribes via speech-to-text [5] and optionally refines the text using an LLM.

**Stage 6: Cloud Upload.** The captured image, 3D position, and annotation are packaged and uploaded to cloud storage.

#### 3.2 Object Retrieval Workflow

Previously documented items can be retrieved at their original 3D locations (Figure 1, right). Upon scanning the marker anchor, the system fetches all records associated with that workspace. Users activate retrieval with a double-pinch gesture: panels appear at each stored position displaying the captured image and voice annotation. A second double-pinch hides all panels, allowing users to navigate the space without visual clutter.

#### 3.3 Implementation

Spatial Inventory runs in Unity on Meta Quest 3 with Meta XR SDK v83, using Passthrough Camera Access for image capture and Depth API for 3D position estimation. A backend server handles SAM2-Large [7] segmentation and Whisper-large-v3 [5] transcription. Multi-view synthesis is performed via API calls to Nano Banana Pro [6].

### 4 EVALUATION

We report technical performance metrics for the core system components.

**System Latency.** Table 1 summarizes processing times for each stage. The critical path from gesture completion to visual feedback is under 0.6 seconds, excluding user time for manual point prompting. Multi-view synthesis relies on Nano Banana Pro, which is only available via API, requiring 12–16 seconds for generation.

**Segmentation and Localization.** SAM2 segmentation performance depends on multiple factors: captured image size, object complexity, point prompt quality, and lighting conditions. In normal daylight indoor environments, we tested 16 common workshop tools (e.g., wire cutters, putty knives, drills) and achieved a

Table 1: System Latency

| Stage                                  | Latency         |
|----------------------------------------|-----------------|
| Gesture + image capture                | ≈95ms           |
| Network round-trip                     | 100–200ms       |
| SAM2 segmentation (RTX 4090)           | 25–75ms         |
| <b>Total (to feedback)</b>             | <b>0.3–0.4s</b> |
| Multi-view synthesis (async)           | 12–16s          |
| Transcription + LLM refinement (async) | ~2s             |

94% first-attempt success rate, with the number of required point prompts ranging from 1 to 8. The depth-ray approach estimates object positions with reasonable accuracy for objects within 2.4 meters; in 32 trials, 29 items appeared at visually correct positions. Errors typically occur when the lasso selection is asymmetric relative to the target object, resulting in 0.3–0.5m depth offset.

### 5 CONCLUSION

We presented Spatial Inventory, an AR system that combines AI-enhanced object capture with a proactive-asynchronous workflow for spatial documentation. Technical evaluation demonstrates the feasibility of integrating foundation models into interactive AR pipelines, achieving sub-second feedback for core operations.

**Limitations.** The system requires cloud connectivity, and segmentation degrades in low-light conditions. Multi-view synthesis may contain hallucinations. Current anchoring assumes fixed object locations, limiting use for portable items. Extended use of head-mounted displays for documentation tasks may not be ideal; we envision this workflow applied to lighter-weight AR glasses in the future.

**Future Work.** Attaching markers directly to objects could enable location-independent retrieval. User studies in authentic workspace settings would validate usability and adoption.

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